Solution, simulation and analysis of the angular dynamics of a free body with a magnetic moment subject to a rotating magnetic field

▾ 📖 **Abstract/Research Context**

This application is a symbolic, numerical, and visual analysis framework for studying the angular dynamics of free or magnetically bounded bodies subjected to rotating magnetic fields. Magnetic bounding by rotating fields is a relatively new phenomenon in which a rigid body with a magnetic moment can be trapped at a precise location by a rotating magnetic field with a gradient, without requiring a local minimum of magnetic potential. The effect is essentially realizable with dipole magnets. In this interaction, the body couples to the field through its magnetic moment, which is fixed in the body frame, and may undergo translational and angular oscillations of typically small amplitude. These oscillations allow the body to find a quasi-stable equilibrium with the driving field and the confinement mechanism which impossible within magnetostatic except using diamagnetism.

This work focuses on the angular part of the dynamics, which is responsible for keeping the body in proper alignment with the rotating field. In a basic model, the angular oscillation consists of a symmetric conical motion of the body's magnetic moment around the axis of the rotating field syncronized with this field. This alignment meschanism can be associated with the phase-lag property of driven oscillators; in this case the phase is held close to π, causing the body's magnetic moment to be antiparallel to the field in the azimuthal components, while parallel alignment occurs in the zenithal components of the magnetic moment and the rotating field.

Since antiparallel alignment results in repulsion and parallel in attraction, the stability in translational degrees of freedom is obtained by the system ability to adjust these opposing effects by variation of the zenithal angle of the body -- which, in turn, corresponds to the amplitude of this driven oscillation as a function of the strength of magnetic torque acting on the body.

As the body moves closer in the field gradient, the stronger field increases the magnetic torque, causing the oscillation amplitude to increase. This corresponds to a larger zenithal angle and makes the antiparallel alignment more pronounced. Consequently, the repulsive component grows faster than the attractive component, producing an effective positive translational stiffness.

A related self-regulating mechanism can also arise from translational oscillations, especially in the radial plane with respect to the rotation axis of the driving field in small circles syncronized by the rotating field. Through the phase-lag response of the driven motion, the displacement of the dipole can become nearly opposite in phase to the effective attractive force. This shifts the dipole (therefore the body) in the field gradient in a way that favors the repulsive component of the magnetic interaction, so that the attractive force is effectively used against itself. The result is an effective positive translational stiffness. Although this displacement may be small, it can still be sufficient to produce a bound state in cases where angular oscillations are constrainted or suppressed.

In the model implemented here, the angular degrees of freedom are described by the three Euler angles θ, ϕ, and ψ. The angle ψ denotes rotation of the body about its magnetic axis, while ϕ and θ describe the azimuthal and zenithal orientation of the magnetic moment in a coordinate system whose z-axis is the rotation axis of the driving field. The magnetic moment is assumed to be fixed in the body frame, while the applied magnetic field rotates with user-defined angular velocity, field inclination, and strength.

The equations of motion are derived using the Euler–Lagrange formalism in the Euler-angle coordinates (θ, ϕ, ψ), for a rigid body with a diagonal moment-of-inertia tensor whose one principal axis coincides with the magnetic axis. Axisymmetric and isotropic cases are also covered as reductions of the general model.

For an axisymmetric body, the equations of motion admit synchronized steady-state solutions in which the vectors entering the dynamics remain time-invariant in the reference frame of the rotating magnetic field. In this rotating-field frame, the explicit time dependence cancels and the equations of motion become autonomous. These equilibria, supported by numerical simulations and experimental observations, provide the operating points for local linearization. The equations of motion are then expanded to first order about the equilibrium state defined by the generalized coordinates and their first derivatives, yielding the state-matrix representation used for eigenvalue analysis.

The eigenvalue analysis exposes the local stability characteristics and natural angular frequencies of the synchronized solution. In the absence of damping, the system can be marginally stable, with eigenvalues whose real parts are zero. In the presence of damping, stable solutions become asymptotically stable when the real parts of all eigenvalues are negative. When the field inclination angle γ is reduced below the required stability threshold, one real part becomes positive, clearly indicating an unstable regime.

The simulations and eigenvalue results support each other in identifying stability limits and reproducing accurately the observed eigenfrequencies. The imaginary parts of the eigenvalues give the natural angular frequencies of the synchronized solution. In the stable regime, two principal eigenfrequencies are typically observed close to the driving frequency; for small oscillation amplitudes, one lies slightly below the driving frequency and the other slightly above it. The separation of these frequency branches is mainly controlled by the field inclination angle γ, while increasing oscillation amplitude tends to shift both frequencies downward. This downward shift can be interpreted as a softening-type nonlinear effect. Through parameter sweeps, the utility makes it possible to observe frequency branching, stability boundaries, damping effects, and the influence of inertia ratios.

Simulations further indicate the existence of excited bounded modes within the stable regime. In these modes, the zenithal angle of the magnetic moment can vary over a large range, and the corresponding state-space trajectories may form cycloidal, rosette-like, looped, or otherwise structured patterns. In damped cases, these trajectories can converge toward attractors or preferred bounded regions in state space, indicating stable excited forms of motion in addition to the basic synchronized equilibrium. The built-in state-variable plots, phase portraits, and polar plots provide a practical way to examine these modes and compare them with regular synchronized motion, beat formation, nonlinear amplitude limitation, and the onset of instability.

Section C analyzes the behavior of the state variables in the presence of damping. In particular, it is shown that, in the absence of electromagnetically induced spin-up torques, $\dot\psi$ converges to $-\omega\cos\theta$. This condition corresponds to a zero z-component of the body's angular velocity in local coordinates and provides a useful reference state for setting up damped simulations.

Direct nonlinear simulations and eigenvalue results support each other in identifying stability limits and reproducing the observed eigenfrequencies. Simulations also reveal an angular instability mechanism when the body has a small non-axisymmetry, which is often encountered in experiments. In such cases, the nonlinear dynamics of the system can use this asymmetry to promote resonant angular modes.

Section E derives Ucar (2021)'s semi-empirical stability criterion from eigenvalue bifurcation and reformulates the formula in terms of the driving-field elevation angle γ. The formula is applicable to isotropic bodies under the condition $\dot\psi = -\omega$ under zero damping and is found to predict stability limits in agreement with both eigenvalue analyses and direct simulations.

A central purpose of the application is to support the investigation of magnetic bound-state dynamics beyond simple numerical simulation. The symbolic derivation of the equations of motion allows the assumptions, coordinate definitions, and model reductions to be inspected directly. The numerical solver enables time-domain simulations, while the animation, plotting, and phase-portrait tools help visualize the physical motion and the evolution of state variables.

The application also includes linearization and eigenvalue analysis tools in Section D. These are useful for studying local stability around synchronized operating states, identifying natural angular frequencies, and comparing predicted small-signal behavior with frequencies observed in full nonlinear simulations. By sweeping parameters such as field rotation rate, field inclination, damping, magnetic torque strength, inertia ratios, and initial conditions, the user can observe how stable, unstable, quasi-synchronized, and transitional regimes emerge.

Therefore, this utility can be used as both a research notebook and an exploratory computational laboratory for angular magnetic dynamics. It may be useful for studying magnetically bounded motion, rotating-field stabilization, resonance-like behavior, damping-dependent equilibrium states, non-axisymmetric instability mechanisms, and the relationship between nonlinear simulations and linearized eigenvalue spectra.

▾ 📖 **Quick Guide**

📘 Quick Guide: Using This Utility

About

This is a customized rigid-body simulation and numerical analysis framework for studying angular motion of a body with a magnetic dipole moment and subjected to an external magnetic field. The body is assumed, in the general case, to have a diagonal moment-of-inertia (MOI) tensor, with its magnetic moment aligned along the body's z-axis and centered at the center of mass. The model supports an external magnetic field of constant strength that can rotate about the z-axis with an elevation angle γ about the xy-plane. Using this scheme, the utility can simulate the angular dynamics of a magnetic body under the magnetic bound-state condition in classical physics (https://doi.org/10.3390/sym13030442) with user-defined parameters. The results can be visualized through animations, and plots that depict the time-evolution of relevant system variables, together with additional figures.

The utility includes derivation of the equations of motion (**Section A**) starting from three Euler rotations and applying the Euler–Lagrange formalism, implemented using the SymPy package. Although the system is externally driven and therefore non-conservative, the Euler–Lagrange approach yields the correct equations of motion because the magnetic torques can be expressed as the spatial derivative of the magnetic potential $V(q,t) = -\vec{m}\cdot\vec{B}(q,t)$ where q denotes the generalized coordinates. Viscous damping is incorporated separately as an external generalized force (torque). All derivations are generated in the `EOMDiag` class, a Python module within this project (see the member list below). Additionally variables which may be useful for further evaluations are also listed there.

Section B summarizes the equations of motion for bodies with diagonal MOI tensor, and with axisymmetry and isotropy as special cases, also provides basic vectors used in derivations.

Section C covers the characteristics of the state variables in the presence of damping, which are particularly useful for setting up equilibrium initial conditions in simulations.

Section D presents the linearization of equations of motion, Jacobians, state-matrix representation and eigenvalue analysis. Using this section, one can calculate and plot eigenvalues of a solution with respect to a user-selected parameter. This way, the stability of a solution and eigenfrequencies can be visualized.

Section E covers derivation of semi-empirical stability criterion originally proposed by Ucar (2021) from eigenvalue bifurcation and reformulates the formula in terms of the driving-field elevation angle γ.

Fast Start

1. **Page Layout**

   Adjust the browser window width so that the interfac fills the page without horizontal scrollbars. The page normally fits within a 1315-pixel width. One can zoom [+][-] from browser control to see the page comfortably.

2. **Choose a preset**

   First-time users may run the default configuration provided in startup or select one of the predefined presets. Each configuration specifies all physical parameters ($I_1,\ I_2,\ I_3,\ mB,\ \omega_B,\ \xi$) as well as the initial values of the state variables ($\theta_0,\ \phi_0,\ \psi_0,\ \dot\theta_0,\ \dot\phi_0,\ \dot\psi_0$) which enter the equations of motion for a body with a diagonal MOI tensor.

3. **Simulate**

   Click on ▶ **Simulate** to generate time-evolution data belong to angular motion of the body, covering angular positions and velocities for each simulation step. In the default simulation settings, steps are set as 2500 steps per second and simulation length is 10 seconds. Step count can be increased to aacomodate to higher speeds of the rotating field. This default setup simulates angular motion of a magnetic body with isotropic moment-of-inertia tensor with default parameters, which can be visualized as a cube, a sphere or a cylinder with appropriate geometry from the scene view. This step may take several seconds to complete. Note that parameters $I_1,\ I_2,\ I_3$ must satisfy $I_1, I_2 < I_3$ in order to represent a physically realizable body.

4. **View plots**

   Expand the **Plots** panel to access graphical representations of the simulation results. By default, only the $\theta(t)$ plot is enabled. Additional time evaluation and phase portrait plots can be displayed using the checkboxes at the top of the panel. Displaying a new plot may take a few seconds. If initial conditions are automatically set for symmetric and stable motion of the body, time evaluation plots may only show residual variations of values, which can be up to 1e-6 for velocity figures and 1e-10 for others.

5. **Generate Eigenvalue Spectrum**

   Expand the **Linearization and Eigenvalue Analysis** panel. Scroll down (or use the shortcut) to the **Eigenvalue Analysis** section. Press the **Plot** button. See real and imaginary parts of eigenvalues in the spectrum. System is stable or quasi-stable when all real parts are zero or negative. Imaginary parts bifurcate in the stable zone. These plots are zoomable from the toolbar at the left and allow precise zooming on the zone boundary using the **Zoom on Bifurcation** button. One can select the sweep parameter and its range to examine how eigenvalues vary with that parameter.

6. **Scene setup**

   Click the **Scene** button, which opens a 3D animation scene at the bottom of the page. Select the shape, the scale and the view angle. View angle can be changed either by mouse dragging (hold the left button and move) or by pressing coordinate buttons (each button cycles through four view angles on repeated presses).

7. **Animate**

   Enter **0.2** into **Span** field and **10** into **Length** field, then click ▶ **Animate**. A progress bar is shown; generation typically completes in under 30 seconds. Upon completion, an embedded video player appears at the bottom of the page.

8. **Download or inspect**

   Generated animation files are stored in the directories 'RESULTS' or 'RESULTS/TEMP' and may be played back or downloaded from JupyterLab interface.

9. **Explore derivation and characteristics of the model**

   - See sections **A** and **B** which expose the physics model used in generation of simulation data.
   - See sections **C, D** and **E** where the stability and characteristics of the dynamics are evaluated.

Step 1: Set Parameters

1. In the **Setup Parameters** column enter:

   - ω_B : Angular velocity of the external field $\vec{B}$
   - I_1, I_2, I_3 : Principal moments of inertia
   - mB : Product of the body's magnetic moment $\vec{m}$ magnitude and field strength of external field $\vec{B}$
   - γ : Tilt angle of the $\vec{B}$ from xy-plane (complement of the zenith angle or elevation angle)
     When $\gamma = 0$, $\vec{B}$ lies entirely in xy-plane. However as small tilt $\gamma_{\min}$ is required for a symmetric solution which generates a z-component of the field and forces the body's magnetic moment be aligned with it. It is found that there is a minimum γ angle ensuring the angular motion of the body symmetric regarding z-axis and can be calculated exactly for an isotropic body ($I_1 = I_2 = I_3$). This formula $\gamma_{\min} = \sin^{-1}\left(\frac{mB}{2I\omega^2}\right)$ is derived from linearized equation of motion of $\ddot\theta$ around the equilibrium point θ_{eq} from the proposed condition $\left.\frac{dF(\theta)}{d\theta}\right|_{\theta=\theta_{eq}} \le I\omega^2$ where F is the right side of the equation of motion about $\ddot\theta$ in a solution where second derivatives of all variables are zero. This solution admits $I\omega^2\sin\theta - mB(\cos(\theta-\gamma)) = 0$ where θ can be calculated as $\theta_{eq} = \tan^{-1}\left(\frac{mB\cos\gamma}{I\omega^2 - mB\sin\gamma}\right)$ when $\phi = \omega t + \pi,\ \psi = -\omega t.$
   - γ_ϵ : One may choose a different γ from $\gamma_{\min}$ and the difference can be set or viewed at the this entry.

2. **Set Initial Angless and Initial Velocitiess**

   If the option **Link parameters for symmetry** is checked, utility will try to set initial conditions for an axisymmetric body ($I1 = I2$) in the second and third entry columns in order to obtain a solution where the second derivatives of all variables are zero. In such a solution the angle θ is set as a positive value less than $\pi/2$ (90°). Beyond that, it can be assumed these parameters do not admit a stable solution. While the equations that calculate initial condition for an equilibrium condition are derived from the axisymmetric solution, their results can be used for small non-axisymmetries, in order to easily evaluate these cases.

   - Once the parameters in **Setup Parameters** are set, one can modify parameters in **Initial Angles** and **Initial Velocities** columns. Setting a parameter in one column may update values in the other column in order to obtain a stable solution when the **link** option is checked.
   - There are two buttons, which set the calculation method, right one for setting ($\dot\phi = \omega,\ \dot\psi = -\omega$) which causes the body to return to its initial position after one cycle of the rotating field without making a turn. The left button sets the calculation mode to ensure $\dot\psi = -\omega\cos\theta$. This option does not satisfy the above condition but sets the z component of the body's angular velocity in the body frame as zero ($\nu_z = 0$). Once this mode is selected, parameters are calculated accordingly.
   - It is possible to set initial condition (angles and velocities) using the state variables at a specified time point of a prior or a saved run. This might be useful for eliminating spurious oscillations caused by initial values do not satisfy equilibrium conditions when the application cannot provide them directly in the defined configuration. To address this, one may first perform a simulation with non-zero damping factor, allowing these oscillations to decay. The state variables at a later time when the system is close to equilibrium, can then be extracted and used as the initial conditions for a second simulation. By entering a time value in the time field (t_0) and pressing the **'Set From Simul. at t_0'** button, the program selects values of state variables ($\theta_0,\ \phi_0,\ \psi_0,\ \dot\theta_0,\ \dot\phi_0,\ \dot\psi_0$) from the closest time point. User may set the check box **%** to reduce $\phi_0,\ \psi_0$ to the $0 - 2\pi$ range. After selecting initial value, one may reset $t_0 = 0$, which also defines the time offset of the simulation since time variable begins at $t = t_0$.

3. **Set Damping & Test Parameterss**

   - **Damping** parameter can be set within ξ entry. It is defined as acceleration per velocity in angular terms and its unit is s^{-1}. It is converted to a torque factor by multiplication with the moment-of-inertia. As explained in Box C, damping forces the body's z-component of angular velocity in local coordinates to zero. This corresponds to the first time derivative of the Euler angle ψ converging to the value $-\omega\ \cos(\theta)$. If one needs to keep $\dot\psi$ constant at a specified value, they can press the button [External $\ddot\psi$], to have the program calculate and apply an external torque in simulation to keep this angular velocity. One can also set this external torque factor manually at the $\ddot\psi$ entry. This parameter is also an acceleration term and converted to torque by multiplying with the body's moment-of-inertia.

Step 2: Generate Simulation Results

- Click ▶ **Simulate** to generate time evaluation of rotation variables and their first derivatives. Calculated values (simulation results) can be saved in the file system as CSV file if defined parameters (associated to an entry defined as Notes) are saved in the system by pressing 'Save' button. Simulation results are automatically saved once the Notes entry is saved. User is asked to add new result, replace the previous result, or not to save results.

Step 3: Inspect Results

- Open the **Plot** section to inspect a set of graphical representations of the simulation results, presented in the time domain and in phase space. The available plots are:
  - $\theta(t)$ and its frequency spectrum
  - Phase of $\phi(t)$ w.r.t. $B(t)$
  - $\dot\phi(t)$
  - $\dot\psi(t)$
  - $\nu_z{}^w, \nu_z{}^b$ (Z component of angular velocity vector in world and in body frames)
  - Kinetic energy $T(t)$ related to angular motion
  - Potential energy $V(t)$ related to magnetic torque
  - Total energy $T(t) + V(t)$
  - Polar plot of (θ, ϕ)
  - Spherical plot of (θ, ϕ) as a projection on a plane
  - Phase portrait of several entities as $(\theta, \dot\theta),\ (\phi - \omega t,\ \dot\phi),\ (\psi, \dot\psi),\ (\theta, T)$ - kinetic, (θ, V) - potential, (θ, H) - total.
- Notes:
  - The magnetic potential can vary between negative and positive values. The zero value corresponds $\vec{m} \perp \vec{B}$, minimum value corresponds to parallel and maximum to antiparallel alignments.
  - Each plot has left side toolbar allowing one to zoom to a specific part of the plot and to navigate. Some plots have measurement boxes at the bottom showing the average and peak to peak amplitude of the signal. These figures can be copied to clipboard by clicking on them.
  - The Sync button at the top right of the plots allows to set the time range of the plot to the time range last zoomed plot. Clicking again the button revert the action.

Step 4: Generate Animation

1. Click **Scene** button to open the visual setup of the animation scene. This includes basic axisymmetric shapes that one can choose, their scales and set the view angle of the scene. Additional rendering parameters can be defined in the **Option** box on the form. The proportion of the body (i.e. radius to height ratio of a cylinder) is automatically calculated respecting I_1/I_3 ratio, moment-of-inertia tensor components.
2. Click ▶ **Animate** to generate an mp4 video of the animation.

Class EOMDiag member variables

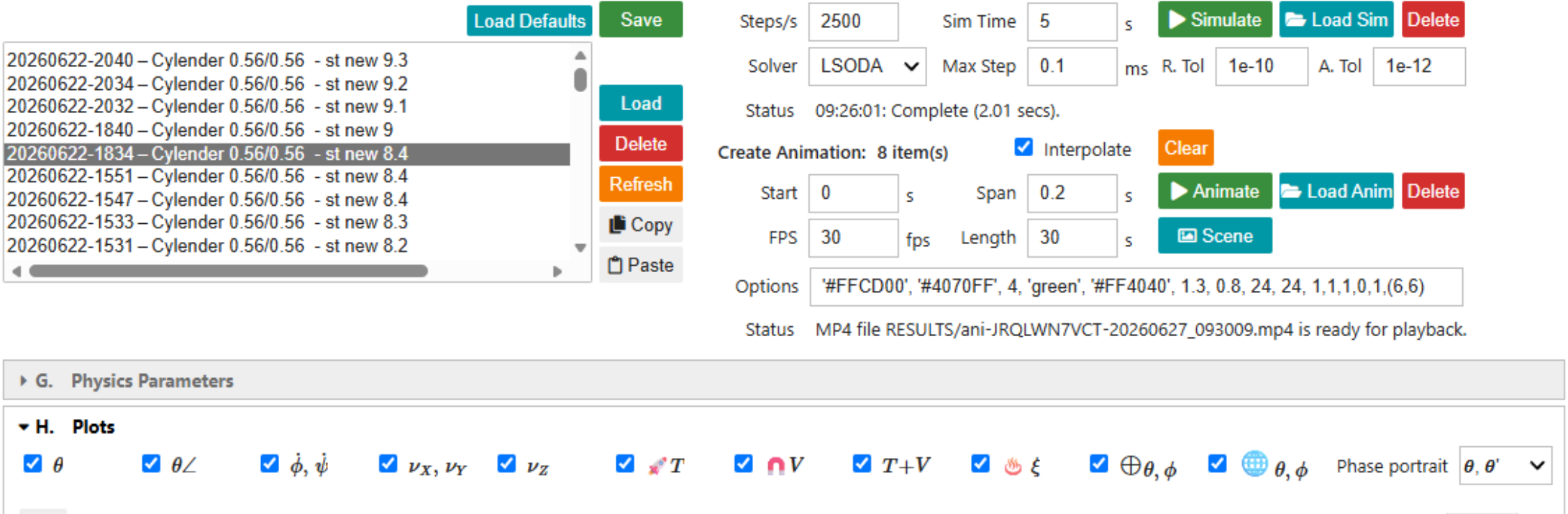
Class variables which can be used for symbolic operations. The instance of the class is defined as **em**.

Variable	Description
`t`	Time
`I1`	Component I_{11} of a diagonal moment-of-inertia tensor
`I2`	Component I_{22} of a diagonal moment-of-inertia tensor
`I3`	Component I_{33} of a diagonal moment-of-inertia tensor
`I`	Isotropic moment-of-inertia scalar
`omega`	Angular velocity ω of the external magnetic field $\vec{B}$
`mB`	mB — product of the magnitudes of body's magnetic moment $\vec{m}$ and field $\vec{B}$
`gamma`	Tilt (elevation) angle γ of the field $\vec{B}$
`xi`	Viscous damping constant ξ
`th`	Euler rotation angle θ (generalized coordinate)
`ph`	Euler rotation angle ϕ (generalized coordinate)
`ps`	Euler rotation angle ψ (generalized coordinate)
`th_d`	$\dot\theta$ — first time derivative of θ (generalized velocity)
`ph_d`	$\dot\phi$ — first time derivative of ϕ (generalized velocity)
`ps_d`	$\dot\psi$ — first derivative of ψ (generalized velocity)
`th_dd`	$\ddot\theta$ — second time derivative of θ (generalized acceleration)
`ph_dd`	$\ddot\phi$ — second time derivative of ϕ (generalized acceleration)
`ps_dd`	$\ddot\psi$ — second time derivative of ψ (generalized acceleration)
`R`	Euler rotation matrix R (ZXZ convention)
`nu_bo`	Body's angular velocity vector $\vec{\nu}_{bo}$ expressed in body-frame
`nu_wo`	Body's angular velocity vector $\vec{\nu}_{wo}$ expressed in world-frame
`um`	Unit vector $\hat{m}$ of the body's magnetic moment
`uB`	Unit vector $\hat{B}$ of the external magnetic field
`I_rot`	Body's moment-of-inertia tensor I_{rot} in world-frame (time dependent)
`T`	Body's rotational kinetic energy T
`V`	Body's rotational potential energy V
`eom`	Equations of motion for a body with diagonal MOI. Indexable as eom[n], n=0,1,2.
`eom_axisym`	Equations of motion for axisymmetric body (indexable similarly)
`eom_iso`	Equations of motion for a body with isotropic MOI (indexable similarly)
`f_th(...)`	RHS function returning $\ddot\theta$ (Euler equation)
`f_ph(...)`	RHS function returning $\ddot\phi$ (Euler equation)
`f_ps(...)`	RHS function returning $\ddot\psi$ (Euler equation)

Variables accesible from other cells

Variable	Description
`t_vals`	Array holding time points of a simulation steps. Indexable as `t_vals[j]` where `j` is a step.
`y_vals`	Array holding state variables of a simulation. Indexable as `y_vals[i, j]` where `j` is the step index and `i` is the variable index where 0 = θ, 1 = ϕ, 2 = ψ, 3 = $\dot\theta$, 4 = $\dot\phi$, 5 = $\dot\psi$
`JEm`	Jacobian matrix (6x3) for axisymmetric bodies.
`JEmIso`	Jacobian matrix (6x3) for isotropic bodies.
`eigvals_all`	Array of Eigenvalues within a spectrum
`eigvecs_all`	Array of Eigenvectors within a spectrum

- ▸ A. Derivation of Equations of Motion
- ▸ B. Basic Definitions and Equations of Motion
- ▸ C. Evaluating State Variables in the Presence of Damping
- ▸ D. Linearization and Eigenvalue Analysis
- ▸ E. Derivation of the Stability Criterion from Eigenvalue Bifurcation



Setup Parameters
- ω_B 100 rev/s
- I_1 5e-7 kg m²
- I_2 5e-7 kg m²
- I_3 3.5e-7 kg m²
- mB 0.036 Nm
- γ_ϵ -2.809774023 deg
- γ 5.932016827375 deg

Initial Angles
- θ_{eq} 111.051016656466 deg
- θ_ϵ 0 deg
- θ_0 111.051016656466 deg
- ϕ_0 85.662740472022 deg
- ψ_0 284.289962679513 deg
- t_0 0 s

Initial Velocities
- $\dot\theta$ -12.805052849373 rev/s
- $\dot\phi$ 10.409620807558 rev/s
- $\dot\psi$ 20.696877870306 rev/s
- $-\dot\psi/\omega$ 1.000000

Damping & Test Params
- $\ddot\psi$ 0.2423 rev/s²
- ξ 0.01 1/s

☐ Link parameters for symmetry

$\nu_{0\,bo}$ = [-12.57496574, -10.01093183, 16.95775170] rev/s

$\nu_{0\,wo}$ = [18.29186142, -14.22916478, 2.97532141] rev/s

- ▸ F. Conversions

Notes: Cylender 0.56/0.56 - st new 8.4

```
20260622-2040 – Cylender 0.56/0.56  - st new 9.3
20260622-2034 – Cylender 0.56/0.56  - st new 9.2
20260622-2032 – Cylender 0.56/0.56  - st new 9.1
20260622-1840 – Cylender 0.56/0.56  - st new 9
20260622-1834 – Cylender 0.56/0.56  - st new 8.4
20260622-1551 – Cylender 0.56/0.56  - st new 8.4
20260622-1547 – Cylender 0.56/0.56  - st new 8.4
20260622-1533 – Cylender 0.56/0.56  - st new 8.3
20260622-1531 – Cylender 0.56/0.56  - st new 8.2
```

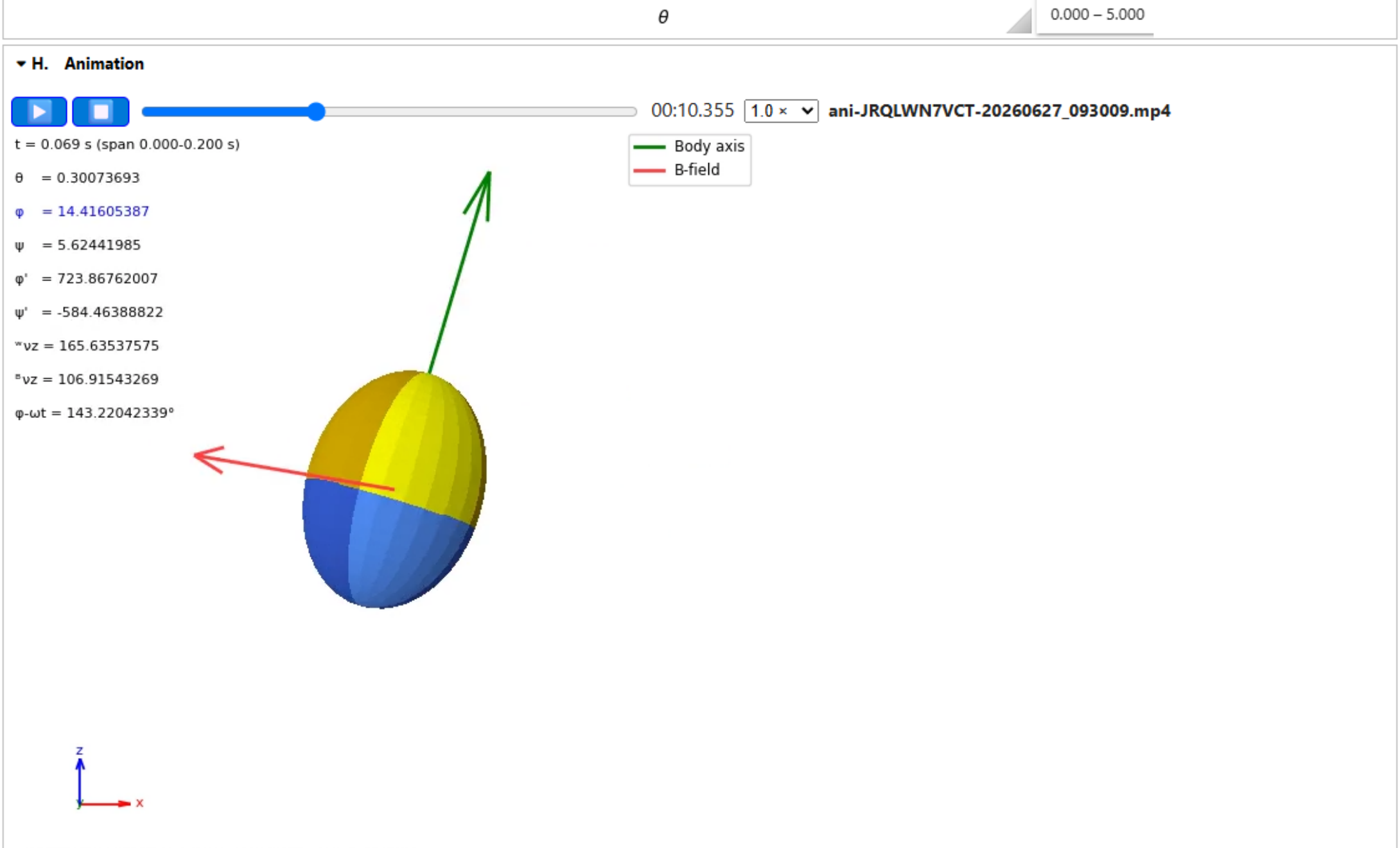
Create Simulation and Plots: 1 item(s) **ID:** JRQLWN7VCT
- Steps/s 2500
- Sim Time 5 s
- Solver LSODA
- Max Step 0.1 ms
- R. Tol 1e-10
- A. Tol 1e-12
- Status 09:26:01: Complete (2.01 secs).

Create Animation: 8 item(s) ☑ Interpolate
- Start 0 s
- Span 0.2 s
- FPS 30 fps
- Length 30 s
- Options '#FFCD00', '#4070FF', 4, 'green', '#FF4040', 1.3, 0.8, 24, 24, 1,1,1,0,1,(6,6)
- Status MP4 file RESULTS/ani-JRQLWN7VCT-20260627_093009.mp4 is ready for playback.

- ▸ G. Physics Parameters
- ▾ H. Plots

☑ θ ☑ θ∠ ☑ φ̇, ψ̇ ☑ νX, νY ☑ νZ ☑ T ☑ V ☑ T+V ☑ ξ ☑ θ, φ ☑ θ, φ Phase portrait θ, θ'
θ(t)
f₂ = 67.7984 Hz 425.9899 rad/s
f₁ = 33.8991 Hz 212.9945 rad/s
θ avg: 1.4365671 pp: 1.9650949
f₀ = 50.8488 Hz 319.4922 rad/s
θ (rad)
Time (sec)
Span=1.09626 s
Frequency Spectrum of θ - (time window: 0.0000 - 1.0963 s)
Magnitude (log)
Frequency (Hz)
Double-click to display peak frequencies in Hz or rad/s
Phase of φ with respect to B field
Phase (deg)
φ₀ avg: -60936.462 pp: 121982.59
φ̇, ψ̇ (Angular velocity of φ and ψ)
Velocity (rad/s)
φ̇ avg: 201.33537 pp: 2226.9776
ψ̇ avg: 7.0822031 pp: 2298.847
Angular velocity X, Y comp. (νx, νy) in body frame f₀ = 1.05579 Hz 6.63374 rad/s
νx avg: -23.476267 pp: 698.42994
νy avg: -2.2512117 pp: 698.08645
Angular velocity Z comp. (νz) in lab and in body frames
LAB νz
BODY νz
νzLA avg: 82.596514 pp: 261.13436
νzBO avg: 106.54885 pp: 0.00031643492
Instantaneous Kinetic Energy
Kinetic Energy (J)
T avg: 0.013469597 pp: 0.028591785
Instantaneous Magnetic Potential Energy
Potential Energy (J)
V avg: 0.0054271162 pp: 0.06746955
Instantaneous Total Energy
Total Energy (J)
avg: 0.019555132 pp: 0.080139541
Instantaneous Viscous Dissipation
Dissipation Power (W)
D avg: 0.00028642094 pp: 0.00057183571
Polar Plot: θ vs φ
Time 0.0000 – 5.0000
Spherical Plot of (θ, φ)
Color DPI 300 Save ...
Phase portrait θ, θ'
☑ Time Arrows
Arrow step 101
☑ Varying Colors
Base color
Color rate 1.00
Reset Parameters
Phase Portrait (θ, θ̇)
Sync
0.000 – 5.000

▾ **H. Animation**



```
t = 0.069 s (span 0.000-0.200 s)
θ    = 0.30073693
φ    = 14.41605387
ψ    = 5.62441985
φ'   = 723.86762007
ψ'   = -584.46388822
ʷνz = 165.63537575
ᴮνz = 106.91543269
φ-ωt = 143.22042339°
```

Source: RESULTS/sim-K5OOC6ELRT-20260621_131102.csv